



Data Mining

Hi!

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Teaching

- [DTM] Big Data and Cloud Platforms (Module 2)
- [DTM] Machine Learning and Data Mining (Module 2)

Research topics

- Big data / database / machine learning
- Precision agriculture and spatio-temporal analytics

BIG (Business Intelligence Group)

-  <https://big-unibo.github.io>
- w/ Prof. Golfarelli, Prof. Gallinucci



Table of Contents and Course Goals

Hands-on *machine learning* (7.5 hours + 7.5 hours with Prof. Gallinucci)

- *Theory, practice*, and *case studies* of the main steps of the CRISP-DM methodology
 - We will have coding labs
 - Python libraries for machine learning (mainly *pandas* and *scikit-learn*)

Goals

1. *Master the right terms and paradigms* behind AI
2. *Develop (=code) your own (simple) project* based on AI
3. *Make technical considerations* about AI
 - “What happens if I do ...?”
4. *Understanding the surrounding AI world*
 - Make decisions with AI
 - Try to make predictions about AI for your company
 - Orient yourself in an extremely dynamic and complex world dominated by AI

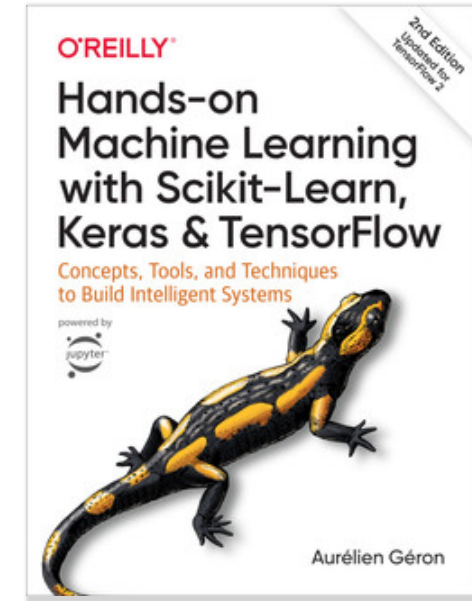
Teaching material

Slides and Python notebooks (to be opened on Google Colab) are sufficient to prepare for the final exam

Download the material from <https://github.com/w4bo/2026-bbs-dm>

However, I suggest you to read the book:

- **Title:** Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow
- **Author:** Aurélien Geron
- Some notes about the book:
 - The book is intended as a support
 - The book is available for free in the library
 - The book is pretty huge (about 800 pages)
 - Carefully check the summary and select our topics
 - If you want, you can read the other parts to take inspiration (not requested for the final exam)

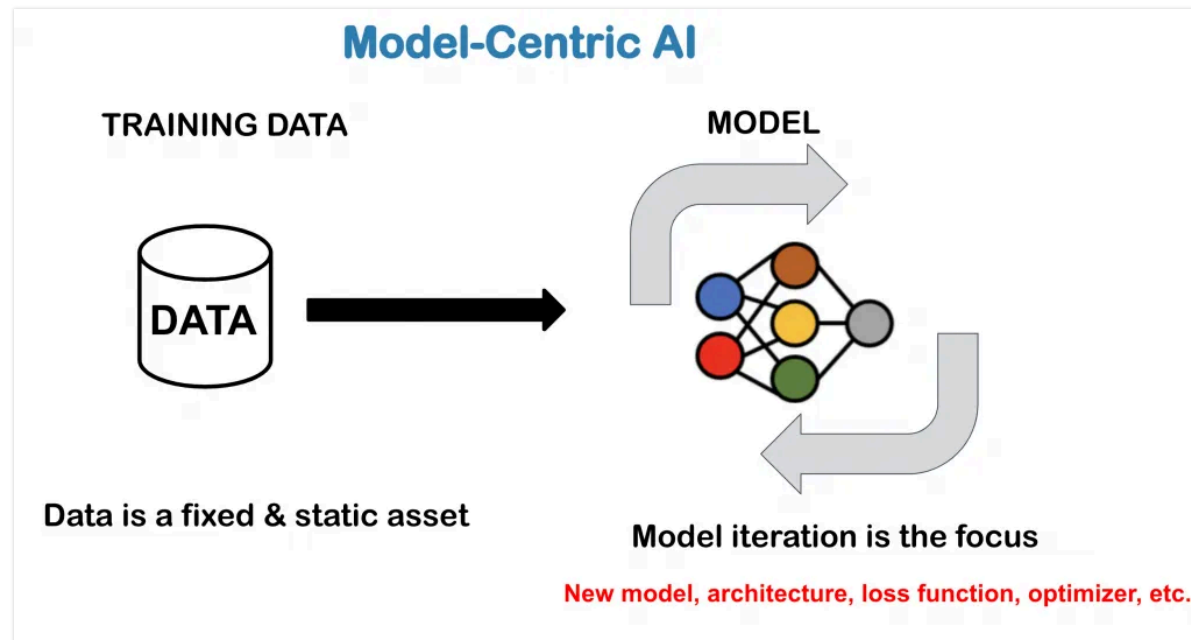


Hands on ML

Interesting news

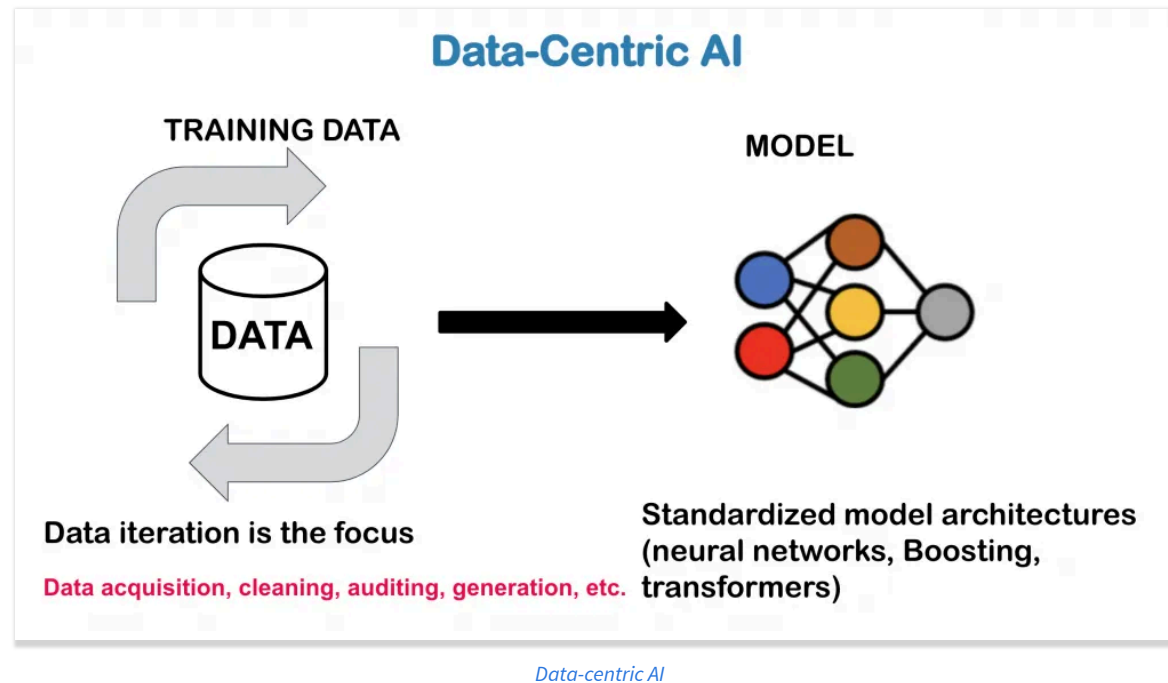
- (April 1, 2025) Ghibli effect: ChatGPT usage hits record after rollout of viral feature
- (January 28, 2025) DeepSeek sparks AI stock selloff; Nvidia posts record market-cap loss
- (January 2, 2025) Nvidia's market value gets \$2 trillion boost in 2024 on AI rally
- [The A.I. Boom Makes Millions for an Unlikely Industry Player](#)
- ... and many others

Model-centric AI vs...



Model-centric AI

... Data-centric AI



Tentative outline

1. (w/ me) From business understanding to data-preprocessing
2. (w/ me) Hands-on machine learning (lab session)
3. (w/ Prof. Gallinucci) Big data pipelines
4. (w/ Prof. Gallinucci) Hands-on big data (lab session)

The focus of my lectures is about *connecting the dots*!

- You have done Machine Learning
- ... and data mining
- ... and others

Let's try to link all the topics and see some interesting (small) case studies!

